VITyarthi | Open Source Software

The Open Source Audit

A Capstone Project for OSS NGMC Course

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1. Introduction

Git is a version control system that helps developers track changes in their codebase over time. It was created by Linus Torvalds in 2005 and has since become one of the most widely used version control systems in the world. Git is a distributed system, meaning that every developer working on a project has a local copy of the entire project history, which makes it easy to collaborate with others. Git is also highly flexible and can be used for a wide range of projects, from small personal projects to large-scale enterprise projects. One of the key features of Git is its ability to create branches, which allows developers to work on different versions of a project without affecting the main codebase. This makes it easy to experiment with new ideas and test them without breaking the main project. Git also has a large community of users and developers who contribute to its development and provide support. The popularity of Git can be seen in the fact that it is used by many large companies, including Google, Facebook, and Microsoft. Git is also widely used in the open source community, with many popular open source projects using it as their version control system. The use of Git has become an essential skill for any developer, and its popularity continues to grow. Git's impact on the software development industry has been significant, and it has changed the way developers work together on projects. Git is a powerful tool that has revolutionized the way we develop software, and its influence can be seen in many areas of the industry. With its flexibility and scalability, Git is an ideal choice for projects of all sizes.

Part A — Origin and Philosophy

A1. Problem this software solves

The problem that Git solves is the need for a reliable and efficient way to manage changes to a codebase over time. Before Git, developers used other version control systems, such as Subversion and CVS, which had limitations and were not as flexible as Git. Git provides a way for developers to track changes to their codebase, including who made the changes, when they were made, and what changes were made. This makes it easy to collaborate with others and to manage different versions of a project. Git also provides a way to create branches, which allows developers to work on different versions of a project without affecting the main codebase. This makes it easy to experiment with new ideas and test

them without breaking the main project. The use of Git has become an essential part of the software development process, and it has changed the way developers work together on projects. Git has also made it easier for developers to work on open source projects, as it provides a way for multiple developers to collaborate on a project without having to worry about version control issues. The benefits of using Git are numerous, and it has become a standard tool in the software development industry. Git's ability to manage changes to a codebase has made it an essential tool for any developer, and its popularity continues to grow. Git has also made it easier for developers to learn from each other, as they can easily share their code and collaborate on projects. The impact of Git on the software development industry has been significant, and it has changed the way developers work together on projects.

A2. License Analysis

Git is released under the GNU General Public License, which is a free software license that allows users to run, study, share, and modify the software. This license gives users the freedom to use Git for any purpose, including commercial use, and to distribute it to others. The GNU General Public License also requires that any modifications to the software must be made available under the same license, which ensures that the software remains free and open. The use of the GNU General Public License has helped to make Git a widely used and popular version control system, as it allows developers to use and modify the software without having to worry about licensing restrictions. The GNU General Public License has also helped to create a community of developers who contribute to the development of Git and provide support for it. The open source license of Git has made it an attractive choice for many developers, as it provides a high degree of flexibility and customization. The GNU General Public License has also helped to ensure that Git remains a free and open source software, which is an important part of its appeal. The license has been widely adopted by many other open source projects, and it has become a standard for open source software. The use of the GNU General Public License has helped to create a large and active community of developers who contribute to the development of Git and provide support for it. The community around Git is very active, and there are many resources available for developers who want to learn more about the software.

A3. Ethics of Open Source

The ethics and community values behind Git are centered around the idea of collaboration and sharing. The developers of Git believe that software should be free and open, and that it should be developed in a collaborative and transparent way. The use of the GNU General Public License reflects this philosophy, as it ensures that the software remains free and open. The community around Git is also very active, with many developers contributing to the development of the software and providing support for it. The community values of Git include a commitment to openness, transparency, and collaboration. The developers of Git believe that software development should be a collaborative process, and that it should be open to anyone who wants to contribute. The community around Git is very diverse, with developers from all over the world contributing to the software. The values of the Git community have helped to create a positive and supportive environment for developers, and they have contributed to the success of the software. The ethics of the Git community are also reflected in the way that the software is developed, with a focus on collaboration and transparency. The use of Git has also helped to promote a culture of collaboration and sharing in the software development industry, and it has changed the way that developers work together on projects. The impact of Git on the software development industry has been significant, and it has helped to create a more collaborative and open community of developers.

Part B — Linux Footprint

Git has a significant footprint in the Linux community, with many Linux developers using it as their version control system. In fact, Git was originally developed for use with the Linux kernel, and it has been used to manage the development of the kernel since 2005. The use of Git has helped to make the development of the Linux kernel more efficient and collaborative, and it has allowed developers to work together more effectively. Git is also widely used in other open source projects, including the Android operating system and the Drupal content management system. The use of Git in these projects has helped to promote a culture of collaboration and sharing, and it has made it easier for developers to contribute to the development of the software. Git is also widely used in the Linux distribution community, with many distributions using it as their version control system. The use of Git has helped to make the development

of Linux distributions more efficient and collaborative, and it has allowed developers to work together more effectively. The footprint of Git in the Linux community is very large, and it continues to grow as more developers adopt the software. The use of Git has also helped to promote a culture of collaboration and sharing in the Linux community, and it has changed the way that developers work together on projects.

Part C — FOSS Ecosystem

The FOSS ecosystem around Git is very large and active, with many developers contributing to the development of the software and providing support for it. The ecosystem includes a wide range of tools and services that are designed to work with Git, including GitHub, GitLab, and Bitbucket. These tools provide a range of features, including version control, issue tracking, and project management. The use of these tools has helped to make the development of software more efficient and collaborative, and it has allowed developers to work together more effectively. The FOSS ecosystem around Git also includes a wide range of open source projects that are built on top of Git, including the Linux kernel and the Android operating system. The use of Git in these projects has helped to promote a culture of collaboration and sharing, and it has made it easier for developers to contribute to the development of the software. The ecosystem around Git is very diverse, with developers from all over the world contributing to the development of the software and providing support for it. The use of Git has also helped to create a large and active community of developers who are passionate about the software and who contribute to its development. The impact of Git on the FOSS ecosystem has been significant, and it has helped to create a more collaborative and open community of developers.

Part D — Open Source vs Proprietary

Git is often compared to proprietary alternatives, such as Subversion and Perforce. These alternatives offer a range of features, including version control, issue tracking, and project management. However, they are often more expensive than Git, and they may not offer the same level of flexibility and customization. Git is also widely used in the open source community, which can make it a more attractive choice for developers who are working on open source projects. The

use of Git has also helped to promote a culture of collaboration and sharing, and it has made it easier for developers to contribute to the development of software. The proprietary alternatives to Git may not offer the same level of community support, which can make it more difficult for developers to get help when they need it. The use of Git has also helped to create a large and active community of developers who are passionate about the software and who contribute to its development. The comparison between Git and proprietary alternatives is often based on the specific needs of the project, and the choice of which one to use will depend on a range of factors, including the size and complexity of the project, and the needs of the developers. The use of Git has been widely adopted in the software development industry, and it continues to grow as more developers adopt the software.

Shell Scripts and Outputs

Script 1: 01-identify.sh

```
#!/bin/bash
#          Author:          Amrit          Raj          Rai          (24MEI10033)

echo
"================================================================================
==="
echo "                       Git AUDIT - SYSTEM IDENTITY                       "
echo
"================================================================================
==="
echo "Linux Distribution: $(cat /etc/os-release | grep PRETTY_NAME | cut -d '='
-f2-                |                tr                -d                '"')"
echo      "Kernel      Version:                                $(uname      -r)"
echo      "Current      User:                                        $(whoami)"
echo        "Home        Directory:                                     $(pwd)"
echo      "System      Uptime:                                        $(uptime)"
echo             "Current            Date/Time:                          $(date)"
echo "--------------------------------------------------------------------------
------"
echo "Message: This system runs on Open Source software, providing freedom to
study,                 change,                 and                 distribute."
echo
"================================================================================
==="
```

When you run this script, it essentially collects and displays basic information about your system. The output will include details such as the Linux distribution you're using, the kernel version, the current user, your home directory, how long the system has been up and running, and the current date and time. You'll also see a message highlighting the fact that the system is built on open source software, which is a key aspect of Linux. Overall, the script provides a quick snapshot of your system's identity, making it a handy tool for getting familiar with your setup or troubleshooting issues.

Script 2: 02-packages.sh

```
#!/bin/bash
#          Author:          Amrit          Raj          Rai          (24MEI10033)

if          [          -f          "/etc/debian_version"          ];          then
    PACKAGE_MANAGER="apt"
    PACKAGE_NAME="git"
else
    PACKAGE_MANAGER="yum"
    PACKAGE_NAME="git"
fi

if   $PACKAGE_MANAGER   list   --installed   |   grep   -q   $PACKAGE_NAME;   then
    echo
"================================================================================
==="
    echo "                      Git AUDIT - PACKAGE INSPECTOR                   "
    echo
"================================================================================
==="
    echo "Status: $PACKAGE_NAME is INSTALLED on this $(cat /etc/os-release | grep
PRETTY_NAME    |    cut    -d    '='    -f2-    |    tr    -d    '"')    system."
    echo "Version: $($PACKAGE_MANAGER show $PACKAGE_NAME | grep Version | cut -d
'                                    '                                    -f2)"
    echo "-----------------------------------------------------------------------
----------"
    echo                 "FOSS                 Philosophy                 Notes:"
    echo " - Git: Git is a free and open source distributed version control
system."
    echo  "  -  Linux:  Linux  is  a  free  and  open-source  operating  system."
    echo   "   -   Vim:   Vim   is   a   free   and   open-source   text   editor."
    echo   "   -   Emacs:   Emacs   is   a   free   and   open-source   text   editor."
    echo
"================================================================================
==="
else
    echo           "$PACKAGE_NAME           is           not           installed."
fi
```

This script checks if the Git package is installed on a Linux system, and if so, it displays the version of Git and some general information about open-source software. It first determines the package manager being used by the system, either apt for Debian-based systems or yum for others. Then, it checks if Git is installed by searching for it in the list of installed packages. If Git is found, the script prints out a formatted message showing that Git is installed, along with its version and some notes about free and open-source software like Git, Linux, Vim, and Emacs. If Git is not installed, it simply displays a message saying that

Git is not installed. The output will display the status and version of Git in a neat format, making it easy to see the results at a glance.

Script 3: 03-auditor.sh

```
#!/bin/bash
#          Author:          Amrit          Raj          Rai          (24MEI10033)

dirs=("/etc" "/var/log" "/usr/bin" "/home" "/root/.git" "/home/$(whoami)/.git")

echo
"================================================================================
==="

echo "                        Git AUDIT - DIRECTORY AUDITOR                    "

echo
"================================================================================
==="
for                    dir                    in                    "${dirs[@]}";
    do
        if           [           -d           "$dir"           ];           then
            echo                           "$dir                           exists"
            echo      "Size:     $(du      -sh      $dir      |      cut      -f1)"
            echo       "Permissions:       $(stat       -c       %a       $dir)"
            echo        "Owner:        $(stat        -c        %U        $dir)"
            echo "-------------------------------------------------------------
------------------"
        else
            echo           "$dir           does           not           exist"
        fi
    done

echo
"================================================================================
==="
```

This bash script, named 03-auditor.sh, is designed to audit a list of specific directories on a Linux system. It checks if each directory exists, and if it does, it provides information about the directory's size, permissions, and owner. The script then displays this information for each directory in the list, which includes system directories like /etc and /var/log, as well as user directories like /home and /root/.git. When you run the script, you can expect to see a neat output in the terminal, with each directory's details displayed one after the other, making it easy to quickly scan and understand the status of these important directories.

Script 4: 04-logs.sh

```
#!/bin/bash
#          Author:          Amrit          Raj          Rai          (24MEI10033)

#       Suggested       log       path       for       Git:       /var/log/git.log

echo
"================================================================================
==="

echo "                        Git AUDIT - LOG FILE ANALYZER                    "

echo
"================================================================================
==="

count=0
while                                read                                line;
    do
        if        echo        "$line"        |        grep        -q        "$2";
            then
                ((count++))
        fi
    done                                  <                                  $1
echo         "Total         occurrences         of         '$2':         $count"

tail          -n          5          $1          |          grep          "$2"

echo
"================================================================================
==="
```

This script is designed to analyze a log file, specifically looking for occurrences of a certain keyword. When you run the script, it reads through the log file line by line, checking if the keyword is present in each line. It keeps track of how many times the keyword appears and displays the total count at the end. The script also shows the last five lines of the log file where the keyword is found, giving you a snapshot of the most recent activity related to that keyword. So, when you run this script, you can expect to see a header with a title, the total count of the keyword, the last five relevant log lines, and a footer with a separator line, all of which helps you quickly understand the keyword's presence in the log file.

Script 5: 05-manifesto.sh

```
#!/bin/bash
#          Author:          Amrit          Raj          Rai          (24MEI10033)

echo
"==============================================================================
=="

echo "                       Git AUDIT - MANIFESTO GENERATOR                   "

echo
"==============================================================================
=="

echo "Please answer the following questions to generate your open-source
manifesto:"

echo -n "What is your name? "
read name

echo -n "What is your favorite open-source project? "
read project

echo -n "Why do you contribute to open-source projects? "
read reason

echo "My name is $name, and I love contributing to $project because $reason." >
$(whoami).txt

echo "Your manifesto has been saved to $(whoami).txt"

echo
"==============================================================================
=="
```

This bash script is designed to generate a personal manifesto for open-source contributors. When you run the script, it will prompt you to answer a few questions, including your name, your favorite open-source project, and why you contribute to such projects. After you provide your answers, the script will create a text file named after your username and write a brief statement to it, summarizing your inputs. For example, if your name is John and you love contributing to Linux, the script might write "My name is John, and I love contributing to Linux because I believe in the power of community-driven software development" to a file named john.txt. The script then informs you that your manifesto has been saved to this file, providing a simple way to express your passion for open-source projects.

Conclusion

In conclusion, Git is a powerful version control system that has revolutionized the way that developers work together on projects. The use of Git has helped to promote a culture of collaboration and sharing, and it has made it easier for developers to contribute to the development of software. The open source license of Git has made it an attractive choice for many developers, as it provides a high degree of flexibility and customization. The community around Git is very active, with many developers contributing to the development of the software and providing support for it. The footprint of Git in the Linux community is very large, and it continues to grow as more developers adopt the software. The FOSS ecosystem around Git is also very large and active, with many developers contributing to the development of the software and providing support for it. The comparison between Git and proprietary alternatives is often based on the specific needs of the project, and the choice of which one to use will depend on a range of factors. Overall, the use of Git has been widely adopted in the software development industry, and it continues to grow as more developers adopt the software. The experience of using Git has been very positive, and it has helped to create a more collaborative and open community of developers. The use of Git has also helped to promote a culture of collaboration and sharing, and it has made it easier for developers to contribute to the development of software. The future of Git looks very bright, and it will continue to play a major role in the software development industry.